

Improving Mechanical Effectiveness During Sprint Acceleration: Practical Recommendations and Guidelines

Dylan Shaun Hicks, MSc,¹ Jake George Schuster, MSc,² Pierre Samozino, PhD,³ and Jean-Benoit Morin, PhD⁴
¹Exercise Science Department, Flinders University, Adelaide, South Australia, Australia; ²Vald Performance and Florida State University Institute of Sports Science and Sports Medicine; ³Univ Savoie Mont Blanc, Laboratoire Interuniversitaire de Biologie de la Motricité, Chambéry, France; ⁴Université Côte d'Azur, LAMHESS, Nice, France

ABSTRACT

Sport scientists and strength and conditioning coaches are showing growing interest in the magnitude, orientation, and application of ground reaction force during acceleration actions in sport, as it can identify the key mechanical determinants of performance. Horizontal force-velocity profiling or sprint profiling helps practitioners understand the capacity of the mechanical force production during the acceleration phase of a sprint. This review examines the methods used in the field for determining horizontal force-velocity (sprint) profiles. It also includes recommendations for practical training methods to address individual force-velocity characteristics, mechanical effectiveness, thereby optimizing acceleration performance.

Address correspondence to Dylan Shaun Hicks, dylan.hicks@flinders.edu.au.

INTRODUCTION

Strength and conditioning coaches are interested in understanding the limitations in mechanical performance during activities involving linear and multidirectional speed. High-speed running (sprinting) is the fundamental component of many team sports and involves 2 key phases: acceleration and maximal velocity (7). The ability to accelerate and reach the highest velocity possible in the shortest period is underpinned by the mechanical components of the neuromuscular system, force, velocity, and power, and specifically the force-velocity (F-v) profile (73). Within the strength and conditioning literature, methods to identify these mechanical components during acceleration have been limited, making it unclear the most appropriate training prescription that should be used to improve these qualities. Therefore, if a resistance training program is designed to enhance sprint acceleration, should strength and conditioning coaches select exercises, which focus on force, velocity, and power, or prioritize one variable over the other?

During the stance phase of a sprint action, a ground reaction force (GRF) is produced, which includes both horizontal and vertical components of the GRF (referred to as horizontal and vertical forces for simplicity), along with the resultant GRF. The stance or contact phase can be divided into braking and propulsive phases in the anteroposterior direction, followed by a flight phase when the limbs are repositioned in the air before contacting the ground again (58). This ongoing exchange of kinematic positions defines sprinting as a ballistic action (58). In comparison with various track and field events where only linear speed is required, in team sports such as Australian rules football and rugby, jumping actions followed by a sprint acceleration in multiple directions are common. These constant changes in velocity require athletes to accelerate

KEY WORDS:

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or decelerate their body mass (24,72) and can include rapid changes in direction to chase down or evade an opponent. Although achieving maximal velocity is important in many team sports (33,39), the ability to accelerate (and decelerate) can be of far greater assistance to an athlete's on-field performance (4,27); therefore, coaches must place a large emphasis on improving this quality.

To accelerate in the horizontal direction in the shortest period, the athlete has to develop the highest net horizontal force possible, averaged across each step during the sprint effort. An individual's ability to perform this task are characteristics of both the mechanical and neuromuscular systems (44), however also influenced by the athlete's technical ability to apply the force and the propulsive impulse (force $\times$ time) produced by the athlete. The constraints of applying force over increasingly shorter periods of ground contact as the athlete moves through the sprint acceleration identify how impulse can affect performance. Acceleration performance will be limited if the impulse is high due to force production occurring over a longer ground contact time. Therefore, the ability to achieve a high net external force applied in the opposite direction to the center of mass displacement, as the running velocity increases, and ground contact decreases, is of primary concern. In many team sports, rapidly changing one's velocity and momentum to evade opponents is crucial (35); however, applying force in a more horizontal direction is a major factor in differentiating between rates of acceleration (13,40,60,61,66).

During the acceleration phase, the ability to apply horizontally oriented force has been shown to be one of the key determining factors to performance (61). This is in contrast to maximal velocity running where Weyand et al. (91) showed that the magnitude of GRF production, oriented vertically over the contact phase, was the limiting factor to performance. Effectively applying lower

limb force in a horizontal direction as velocity increases has been referred to as mechanical effectiveness (75). This mechanical description is underpinned by the force applied by the athlete across the acceleration effort and describes the ratio of the net horizontal component and resultant GRF across the acceleration (61). One "simple" macroscopic method used to determine mechanical effectiveness across a sprint acceleration is horizontal F-v profiling, also known as sprint profiling. Across a sprint acceleration effort, sprint profiling models the step-averaged mechanical outputs (force, velocity, and power) in the horizontal direction. This innovative method provides a detailed "roadmap" for understanding the mechanical components underpinning acceleration. As a means of accurately assessing the horizontal force produced by an athlete, sprint profiling assists coaches to calculate the degree of horizontally directed force applied over any distance or velocity across the sprint effort (62). It also identifies the athlete's mechanical strengths and weaknesses when accelerating, specifically their ability to apply horizontal force and accelerate toward maximal velocity.

Sprint profiling helps coaches and athletes understand the F-v and power-velocity (P-v) relationships, along with how horizontal force production capacity changes across the acceleration, and provides a global view of the likely morphological and neuromuscular properties involved (21). Furthermore, when attempting to understand the mechanical variables that contribute to acceleration performance, it raises the question of whether the conventional approach of manually or electronically timing a 40-yard sprint should be used in conjunction with the more in-depth sprint profiling. Moreover, can this information be effectively used to individualize a resistance training program to target the mechanical strengths and weaknesses of the athlete, thereby improving performance? Additional details provided

by mechanical sprint profiling including power and force orientation provide practitioners with superior means to objectively evaluate, effect, and monitor sprint qualities.

Although sprinting is the most specific and highest velocity training method used to improve an athlete's linear speed, strength and conditioning coaches will often look to other resistance training methods to complement speed training. These methods are used to further elicit adaptations to F-v characteristics and to address various mechanical qualities contributing to performance. The selection of exercises to improve physical performance in a sport should be based on factors that demonstrate the highest transfer to that sport. Because horizontal and vertical components of the GRF are produced while accelerating, yet in different magnitudes, there is often conjecture on where the focus should be placed from an exercise selection perspective when producing force: in the horizontal or vertical direction? Two concepts, which will be discussed in this review regarding exercise selection, are dynamic correspondence (90) and the force-vector theory. These concepts describe that the biomechanics, force production and orientation, and velocity of training movements should be similar to those used in the athlete's sport. Both concepts provide a framework for exercise selection. Yet, when selecting resistance training exercises to improve acceleration performance, should strength and conditioning coaches select exercises based on specificity to the sprint action or maintain a broad approach when attempting to change F-v characteristics?

This review aims to provide background information on the F-v relationship, determinants and biomechanics of acceleration performance, and sprint profiling, as well as discussing exercise selection and training programs for improving athletes' mechanical effectiveness during

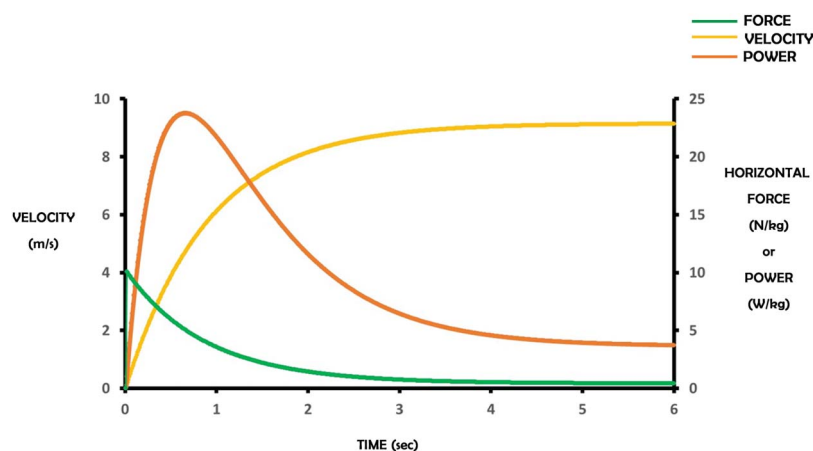


Figure 1. Changes in horizontal force and power as running velocity increases.

acceleration. The practical recommendations in this review could be used to address F-v characteristics and horizontal force application and devise individualized training programs for teams and individual sport athletes.

DETERMINANTS OF FORCE AND VELOCITY

Mechanical variables such as force and velocity play a vital role in ballistic activities such as sprinting and determine overall neuromuscular performance (75). However, these variables are in a sense limiting given that the force produced and the

shortening velocity of skeletal muscle are constrained by morphological factors such as fiber type, fascicle length, pennation angle, and neural mechanisms such as motor unit recruitment and intramuscular coordination (21). Each of these variables has a direct effect on the ability of skeletal muscle to exert maximal power (P_{MAX}). High-power outputs are considered critical performance characteristics for success and will often differentiate between ability levels in sport (84). Practitioners have long argued that athletes should be training at loads, which maximize

power (18,46,83,92); however, other investigators have suggested that loads, which are above and below optimal load, develop P_{MAX} to a greater degree (37,52) warranting further exploration to determine whether an “optimal load” exists and leads to comparatively greater training-induced improvements. It has been shown that ballistic activities are determined by the P_{MAX} of the lower limbs and impulse but are also strongly influenced by the individual’s F-v capabilities, which is also known as the F-v profile (76). Training status and relative strength also influence force expression, and therefore, evaluations of F-v profiles should be highly standardized to maximize reliability of data (38,43). Understanding an athlete’s strengths and weaknesses in terms of their mechanical output assists a strength and conditioning coach to devise an appropriate training program based on the specific needs of the athlete’s F-v profile.

BIOMECHANICAL DETERMINANTS OF SPRINT ACCELERATION

Newtonian laws show that sprint acceleration in a forward direction is determined by the horizontal and vertical components of the resultant GRF, the horizontal and vertical impulse, and the displacement of the center of mass (CoM) (64). Force and impulse are vector quantities, which include direction and magnitude, and depend on the phase of the sprint action, along with the position of the athlete’s body. These vectors are oriented either horizontally (mainly anteroposterior) or vertically. When starting from zero velocity, the impulse will be a combination of force applied over longer ground contacts, and as velocity increases, the time in which force can be applied reduces, therefore making quality force application at ground contact critical. Although net horizontal force determines the rate of acceleration (70,75), the impulse-momentum relationship governs the time in which force is applied; it has been shown that this factor accounts for slow or fast rates of acceleration, where

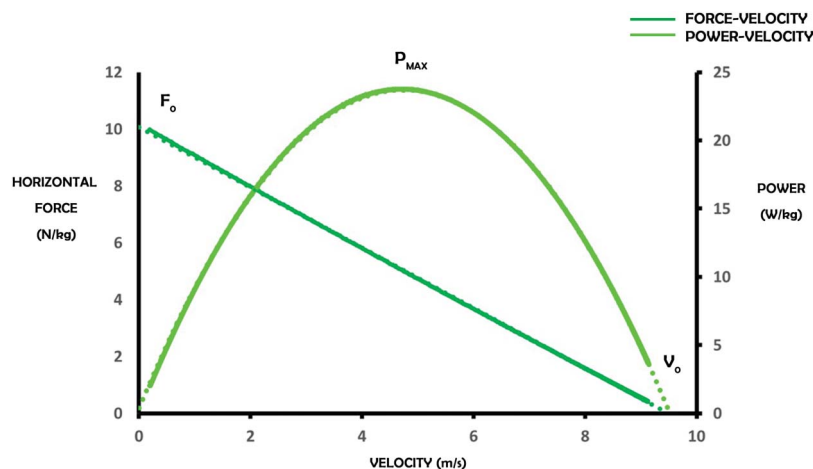


Figure 2. Mechanical output across a sprint acceleration effort. These variables identify the current performance output of the athlete and the mechanical limits of the neuromuscular system: theoretical maximal force (F_0), theoretical maximal velocity (V_0), and maximal power (P_{MAX}) in the horizontal direction.

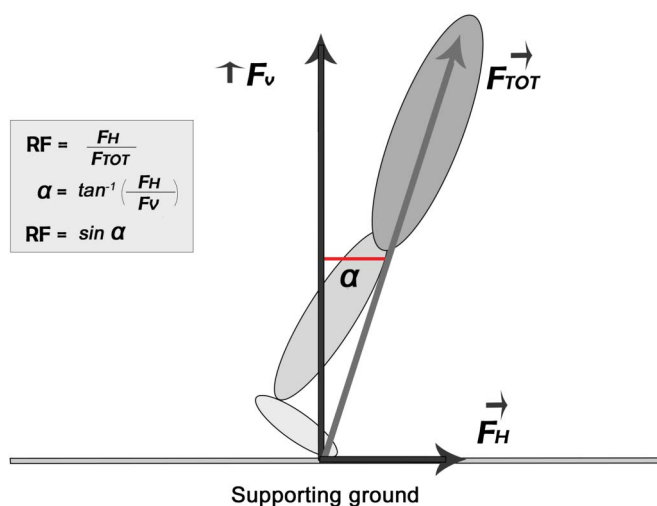


Figure 3. A representation of lower limb ratio of forces, net positive horizontal (F_H) divided by total force (F_{TOT} , which includes the vertical component). The forward orientation of the total GRF vector is represented by the angle α . GRF = ground reaction force.

shorter contact times beget the need for increased force expression. Hunter et al. (40) identified in a series of 25-m sprints that the greatest variance (61%) occurred with the horizontal impulse measured at the 16-m mark. Morin et al. (64) supported this view and the argument that the fastest sprinters were able to produce greater net horizontal impulse compared with their subelite counterparts. Also, of importance, it was shown that the faster sprinters maintained this impulse across the duration of the sprint acceleration as

velocity was increasing and ground contact was decreasing. This was critical to performance.

The way in which a GRF is oriented is key to the acceleration performance or maximal velocity achieved in sprinting (6). Emphasis must be placed on maximizing and orienting horizontal (anteposterior) force application during acceleration because the speed runners ultimately attain specifically correlates with the magnitude of the propulsive force (and time over which it is

applied) at the start of the effort, along with the successive strides during acceleration (12,78,94). It has been shown that elite sprinters produce higher net horizontal force and impulse with each step at any given velocity, which allows them to attain higher velocities than their subelite counterparts (60,61,70). Although the orientation of force is superior in elite sprinters, their training history and kinematics mean that they are also more effective at transferring force into the ground. Such technical skills are also derived from specific neuromuscular properties including the structural integrity of the muscle and tendon (60,70).

The position of the athlete's body when sprinting, whether accelerating or at maximal velocity, influences application and orientation of force (48). Positioning the overall body (not only the trunk-head segments) in an inclined position in relation to the ground makes it possible to achieve a more propulsive resultant GRF (8,12,48). Whereas, when an athlete is sprinting at maximal velocity in an upright position, a greater reliance is placed on achieving high GRF with a vertical orientation to limit time spent on the ground, thereby reducing deceleration (11,91). Directing the resultant GRF in a more forward or horizontally oriented direction is more important during the acceleration phase of a sprint compared with the overall magnitude of force applied to the ground, and therefore, this component is critical to focus on during training (13,60,61,70). Colyer et al. (13) showed that sprinters, compared with soccer players, exhibit more horizontally directed force during the late braking phase and early propulsive phase, allowing them to accelerate to higher velocities; this was a key difference between athlete groups. Orientation of force is also affected by the touch-down or ground contact distance in reference to the body CoM on ground contact (7). During this contact in early stance phase, maintaining a stiff ankle increases the resultant GRF and

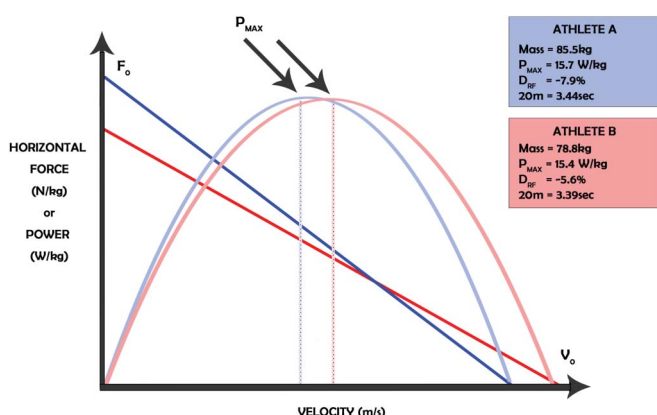


Figure 4. Horizontal force-velocity-power profiles for 2 athletes. Both athletes display similar maximal horizontal power outputs and sprint times, yet different theoretical maximal force and velocity values (see slope).

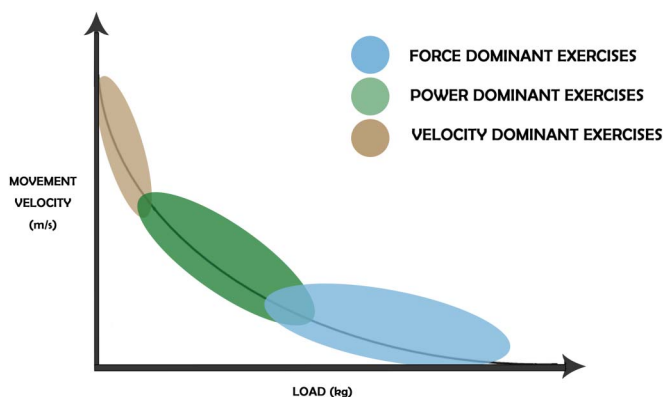


Figure 5. Resistance training categories across the force-velocity (load-velocity) spectrum used to modify the mechanical variables or individualize the F-v profile.

momentum due to the impulse and subsequent horizontal velocity achieved (9). Therefore, assessing and diagnosing the way in which athletes apply horizontal force during acceleration has important ramifications for attaining the best possible sporting performance.

HORIZONTAL FORCE-VELOCITY PROFILING

Horizontal F-v profiling (sprint profiling) is an assessment and diagnostic tool that examines the key characteristics of F-v and P-v relationships in sprint actions; its main focus is on the acceleration phase (26,62). These relationships define the changes in

propulsive force and horizontal power when running velocity increases (Figure 1) (62). Using a series of timing gates or a radar device, as well as biomechanical modeling derived from speed-time data (75), it is possible to calculate horizontal force, velocity, and power as the athlete accelerates. This information describes the current mechanical output from the athlete, along with the mechanical limits of the neuromuscular system while accelerating. Limits include theoretical maximal horizontal force at null velocity (F_0), theoretical maximal horizontal velocity until which force can be produced (V_0), and the maximal power produced in a horizontal direction

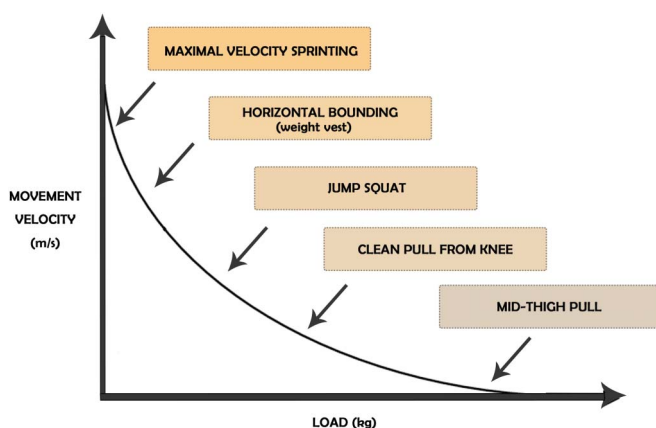


Figure 6. A selection of exercises across the force-velocity (load-velocity) spectrum will be prescribed to each athlete depending on their level of mechanical effectiveness across the acceleration phase.

(P_{MAX}) (Figure 2) (75). Over the duration of a sprint acceleration, Morin et al. (61) use the term ratio of forces (RF)—which describes the horizontal (anteroposterior) component of the GRF (F_H) vector as a percentage of the total GRF (F_{TOT}) vector (Figure 3). This ratio identifies the technical ability an athlete may or may not possess to orient force horizontally while accelerating. Because orientation of the force is more important than its magnitude, understanding the force ratio is critical. From these data, the mechanical effectiveness of applying force ($RF\% = F_H / F_{TOT}$) at each step can be determined. The higher the $RF\%$, the more horizontal orientation of the GRF has been achieved. Mechanical effectiveness is important for determining the athlete's ratio of decreases in force (D_{RF}) with increasing velocity (62), which describes how force orientation changes from more horizontal to vertical. Morin et al. (61) state that even if F_{TOT} is similar in 2 athletes, the $RF\%$ can identify mechanical differences including weaknesses, which can then be targeted with training interventions. Quantifying individuals' mechanical effectiveness during sprint acceleration means it is possible to determine differences between performers but also to establish a biomechanical link between profile and sprint performance (74).

Field-based sprint profiling (63,70,75) using inverse dynamics, a computation method of calculating forces from kinematics of a body, is a highly reliable process that has been evaluated against gold standard laboratory-based (65,67,75) tests using inbuilt force plate systems. Field-based methods of profiling, referred to by Samozino et al. (75) as a simple method, are a practical process needing limited technology and equipment to determine an individual's mechanical profile and assess the P_{MAX} the neuromuscular system is able to achieve during the acceleration phase. Sprint profiling assists coaches to identify the specific interventions required to improve acceleration and determine whether training should be directed at increasing P_{MAX} by

improving the horizontal force produced at low velocity, (force quality), horizontal force at high velocity (velocity quality), or by training at optimal load (maximal power). Sprint profiling can provide some unique findings, given that it is able to distinguish between athletes independently of P_{MAX} values or sprint times. Although time is the critical factor in sprint acceleration, 2 athletes may achieve similar acceleration times and P_{MAX} values over a given distance yet with very different slopes and mechanical characteristics to their F-v profiles (Figure 4). This is connected to an athlete's ability to have a different combination (described as balance or imbalance by Morin and Samozino (62)) between force and velocity (force dominant or velocity dominant), which is also related to their mechanical effectiveness for the duration of sprint

acceleration (62). In comparison to generic training programs where the focus is on improving absolute force and sprint times, sprint profiling provides a specific guide for identifying and targeting the athlete's strengths or weaknesses to improve their acceleration performance. This approach has been explored with elite female athletes in Rugby sevens (80) and team handball players (71), where individual speed training programs based on data from sprint profiles showed varying levels of effectiveness depending on how the sprint profiles were interpreted and how training loads were implemented. Morin and Samozino (62) provided a written explanation about the process of optimizing F-v profiles, but information about practical sprint and resistance training interventions that may have assisted coaches was limited.

PRACTICAL APPLICATIONS AND GUIDELINES

The mechanical determinants and variables seen in profiles such as force, velocity, and power are susceptible to the demands imposed on the body, and key neuromuscular adaptations can occur as a result of prescribing specific exercises (87). This provides scope for a strength and conditioning coach to improve acceleration performance by selecting exercises and loads, which mostly target specific areas on the theoretical F-v spectrum and practical load-velocity spectrum: force, velocity, or power (Figure 5). The resistance training exercises used in most sports are traditionally prescribed off characteristics across the F-v spectrum and the load-velocity (and thus force) context they induce within the movement. Examples of exercises that span this spectrum are detailed in Figure 6.

Team sport weekly microcycle						
MESOCYCLE	Preparation Phase	MICROCYCLE	5	CLUB / SPORT	U21 WOMEN'S	SOCCER
SUNDAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
19-May-19	20-May-19	21-May-19	22-May-19	23-May-19	24-May-19	25-May-19
Rest Day	Pitch C	Pitch B	Pool	Pitch B	Pitch A	Home Stadium
Recovery	Technical + Tactical	Acceleration + S&C	Aerobic Capacity	Tech/Tact. + S&C	Positional / Individual	Accel. MxV + SSG
INTENSITY	INTENSITY	INTENSITY	INTENSITY	INTENSITY	INTENSITY	INTENSITY
LOW	MEDIUM	HARD	LOW	MEDIUM	MEDIUM	HARD
Predicted sRPE - 1	Predicted sRPE - 6	Predicted sRPE - 8	Predicted sRPE - 3	Predicted sRPE - 7	Predicted sRPE - 5	Predicted sRPE - 8
ANCILLARY	ANCILLARY	ANCILLARY	ANCILLARY	ANCILLARY	ANCILLARY	ANCILLARY
NIL	MOBILITY & ACTIVATION	MOBILITY & ACTIVATION	NIL	MOBILITY & ACTIVATION	SOFT TISSUE WORK	MOBILITY & ACTIVATION
TRAINING	TRAINING	TRAINING	TRAINING	TRAINING	TRAINING	TRAINING
Players day off	Attackers, defenders and neutral support on section of 1/2 pitch. Focus on movement off ball and manipulation of defenders. Neutral players cycle in on direction of coach.	Linear > Static: 2 x 10, 15, 20. Rolling: 2 x 10, 15, 20. Curved > Rolling: 2 x 10, 15, 20	Dynamic warmup in pool in shallows. 2 x Session A in 25m pool using fins. 1 x Session B performing DWR + power series against pool wall.	Formation A & B played against reserve grade. 20min blocks. 5min period between blocks.	Attackers, defenders split and matched up with reserves. Finishing drills for attackers. 2 v 1, 3 v 1, 3 v 2 for defenders. GKs Pitch C.	Linear > Static: 4 x 10, Rolling: 4 x 10, Curved > Rolling: 4 x 10, Max velocity wicket runs 4 x 20
	Attackers focus. Final third attacking principles + set piece design and implementation.	Clean Pull (4 x 2 x 82.5%), Back Squat (4 x 2 x 87.5%), Mid-Thigh Pull (3 x 5 x 110%)		Resisted sprinting (2 x 4 x 20, load to reduce mxv 30%), Prowler March (2 x 20, 30, 40m @ 120% BW), Hip Thrusts (5 x 5 x 82.5%)	Conditioning block for RTP, reserves and optional Top-up for players recently back from injury.	Tactical Small-Sided Games (1/2 Pitch). Full-sized goals. 6 v 6 incl. GK. 2-touch play max. Rolling team on 7mins.
POST WORKOUT	POST WORKOUT	POST WORKOUT	POST WORKOUT	POST WORKOUT	POST WORKOUT	POST WORKOUT
	CORE-SERIES	CWI	STRETCH SERIES	CORE-SERIES	STRETCH SERIES	CWI
NOTES	NOTES	NOTES	NOTES	NOTES	NOTES	NOTES
Send wellness details to coaching staff.	Bring field and track footwear for tomorrow's session.	Meet at pool in morning (bathers, fins, goggles)	Bring field and track footwear for tomorrow's session.	Optional physio/chiro appl.	Bring field and track footwear for tomorrow's session.	

Figure 7. A weekly microcycle for a team sport detailing the integration of the technical and tactical sport focus, injury prevention, recovery, and resistance training program (individualized to athlete sprint profile).

Table 1
Exercises to improve maximal force production

Exercise	% 1RM/Load
Back squat	> 85%
Kettlebell swing	> 85%
Romanian deadlift	> 85%
Trap bar deadlift	> 85%
Hip thrust	> 85%
Midhigh pull	> 100% clean
Clean pull from knee	> 100% clean
Rack pull	> 100% deadlift
Prowler march	Up to 150% BW
Resisted sprinting	Up to 100% BW

BW = bodyweight; 1RM = 1 repetition maximum.

Resistance such as an athlete's bodyweight against gravity or external loads is a way to set the velocity at which the maximum effort will occur and indicate the production of force that is possible. Studies that have used resistance training (including vertically and horizontally oriented exercises) to improve sprint performance have included high-force/low-velocity exercises; force-dominant (2,37,47), low-force/high-velocity exercises; velocity-dominant (2,5,10,23,47,52) and optimal load exercises; and power-dominant exercises (27,37), suggesting that the load, orientation, and mechanical focus may elicit different adaptations to the performance.

Categorization of resistance training exercises is useful to understand how adaptations to the profile will affect physical performance. Force-dominant exercises are aimed at improving the force applied at very low velocities. In regard to sprinting, these exercises focus on the athlete's ability to overcome inertia at the start of the sprint acceleration and effectively apply force in a backward direction, be it by improving the capacity of the lower limb force produced or peak mechanical effectiveness. Velocity-dominant exercises are aimed at

improving the application of force at high velocities to enhance the athlete's ability to maintain force application as velocity increases. This can be achieved by improving the lower limb force production at high velocities and/or by improving the orientation of force and maintaining the highest possible mechanical effectiveness despite the increase in velocity. Power-dominant exercises aim to improve the force applied at moderate velocities, that is, at close to half of the theoretical maximal velocity (28,29). These exercises stimulate the athlete's ability to produce greater P_{MAX} output during the sprint acceleration and, when prioritized as interventions within a training program and periodized appropriately, can be effective in enhancing performance. The aim of selecting exercises across the F-v spectrum is to target the variable contributing to the current level of F-v imbalance, thereby improving the athlete's overall mechanical effectiveness across the sprint acceleration.

It is advisable that when selecting resistance training exercises, they demonstrate transfer to movement task and enhance various characteristics that contribute to sprint acceleration. Sprinting is performed on a horizontal training axis

(sagittal plane); therefore, it may seem intuitive to focus on exercises that develop force in the same direction (72,95), known as the "force-vector theory" (14). Using exercises that allow athletes to apply force in the same direction (vector-magnitude and force) as that which occurs in the sport task may suggest a greater transference effect (93) or dynamic correspondence (90) due to similar overall biomechanical characteristics. Using these concepts as an example, volleyball or basketball players often express movements vertically and therefore should address the F-v spectrum by prioritizing exercises that have a vertical force orientation. In comparison, American football players, rugby players, and sprinters, who predominantly express movement through linear locomotion, would be recommended to prioritize horizontally oriented exercises (50,93). Although conjecture surrounds the application of the force-vector theory (34) (see the Limitations section), a thorough understanding of the kinetics and kinematics of the movement task is essential when designing a resistance training programs.

When designing and programming training sessions to improve an athlete's horizontal profile, strength and conditioning coaches need to appropriately periodize resistance training-focused sessions into the weekly sport training program. The structure of a training week in a team sport must primarily focus on the tactical and technical elements of the sport and then prioritize other modalities such as injury prevention, recovery modalities, and resistance training (Figure 7). For optimal F-v adaptations, resistance training should occur over the course of several mesocycles (30) or until the F-v profile has been reassessed and adaptations that contribute to improved P_{MAX} and/or a reduction in F-v imbalance are evident. Continual assessments of the vertical profile (jumping) to determine whether F-v adaptations had occurred were regarded as critically important within a recent study (44). Depending on the level of F-v imbalance revealed in the profile, some or all of the exercises



Figure 8. Resisted sprint training using a sled at 85% bodyweight (59).

identified in the following sections could be integrated into a weekly microcycle, ensuring a minimum 48-hour recovery period between high-intensity days. This is necessary to limit the level of residual fatigue before the athlete embarks on the next training session. Understanding the training phase and how this may affect the general or specific nature of exercise intensity and selection is also a critical factor in team sports (30). Schuster et al. (79) explored these concepts in their recommendations for physical preparation with Rugby 7 athletes, where a weekly combination of high (2 sessions), medium (2 sessions), and low (2 sessions) intensity sessions, including

strength and conditioning, rugby-specific training, and recovery sessions, were cycled across a week to optimize performance during the preparation block leading into competition.

Recommendations about addressing the F-v imbalance and mechanical effectiveness in sprint acceleration through targeted resistance training programs directed across the F-v spectrum are detailed in the following sections.

IMPROVING FORCE PRODUCTION AT LOW VELOCITIES

Athletes with physiological and perhaps technical qualities that limit their ability to apply a high amount of horizontal

force at low velocities are at a disadvantage in many on-field competitive situations. This will be evident early on during the sprint effort with their inability to apply enough horizontal force, thereby reducing their horizontal impulse. In turn, this will compromise the overall velocity that is achieved as this is determined by the athlete's ability to accelerate to this speed. To improve the force produced at low velocities, the prescribed sprint and resistance training needs to include movements that focus on the right-hand side of the F-v spectrum, where force is applied against a heavy external resistance, $>85\%$ 1 repetition maximum (1RM) (86), and targets maximal strength qualities (Table 1).

Exercises that target maximal or absolute strength and specifically improve the force applied in a horizontal direction include heavy sled pulls, resisted sprinting (Figure 8), and prowler marches (Figure 9). Horizontally oriented exercises at these loads will encourage force application in the same direction as what occurs during the acceleration phase of a sprint. Although maximal strength exercises may only be specific to early acceleration, several studies (3,86,87) that focused on the right-hand side of the F-v spectrum noted the crucial role strength plays in providing the foundation to improving maximal power, highlighting its importance for potentially improving other aspects of the F-v spectrum. Table 2 identifies 2 resistance training sessions, which could be performed across 1 week, including both horizontally and vertically oriented exercises. These exercises and the associated sets, repetitions, and loads are programmed to improve the maximal force produced at low velocities.

IMPROVING FORCE PRODUCTION AT HIGH VELOCITIES

Some athletes are capable of high levels of force at low velocities but cannot sustain it as their acceleration increases. This often leads to a rapid decrease in the RF (D_{RF}) as the athlete approaches top speed. Analysis of the profile shows that it is likely that during



Figure 9. Resisted sprint training using a prowler sled to march at 140% bodyweight.

Table 2
Force production at low velocity

Exercise	Week 1		Week 2		Week 3		Week 4	
	Volume	Load	Volume	Load	Volume	Load	Volume	Load
Day 1— horizontal orientation								
Resisted sprinting	2 × 4 × 10 m	Load that restricts to <30% of maximal velocity	2 × 4 × 20 m	Load that restricts to <30% of maximal velocity	2 × 5 × 20 m	Load that restricts to <30% of maximal velocity	2 × 3 × 20 m	Load that restricts to <30% of maximal velocity
Prowler march	2 × 20, 30, 40 m	120% of BW	2 × 20, 30, 40 m	130% of BW	2 × 20, 30, 40 m	140% of BW	2 × 20, 30, 40 m	120% of BW
Hip thrusts	5 × 5	82.5%	5 × 5	87.5%	5 × 5	92.5%	5 × 5	85%
Day 2— vertical orientation								
Clean pull	4 × 2	82.5%	6 × 2	87.5%	12 × 1	92.5%	4 × 2	85%
Back squat	3 × 5	87.5%	3 × 3	90%	3 × 3	92.5%	3 × 5	85%
Midhigh pull (% based off clean)	3 × 5	110%	3 × 3	120%	3 × 3	130%	3 × 3	110%
BW = bodyweight, % based off 1-repetition maximum in exercise.								

acceleration, the athlete will be losing their ability to apply and orient horizontal force too early, which during acceleration corresponds to early changes in body position from approximately more horizontal to vertical. Although the production of high (mostly vertically oriented) force is vital at maximal velocity (11,12,91), the speed attained will be limited because of a rapid decrease in the RF. This has a direct impact in sporting activities such as rugby when players try to outrun their opponents when making for the try line. Sprinters face the same problem when they need to maintain acceleration for a longer duration and reach higher velocities in a 100-m sprint. Improving sprint acceleration performance over longer distances and maintaining a high ratio of horizontal-to-resultant force at

increasing velocities require exercises that focus on characteristics from the left-hand side of the F-v spectrum, along with improved inter- and intramuscular coordination properties. Exercises demanding high velocity are generally those that require high rates of force (Table 3).

The F-v spectrum suggests that the smallest load the human body can work against is the force of gravity on body mass such as when performing a vertical jump. However, research suggests that even this load may be too great to affect the velocity portion of the F-v spectrum (51). Assisted vertical jumps (Figure 10), using elastic bands, are one method that has been used to deload or negatively load an athlete's body mass by reducing the effects of gravity on the body (51). Markovic and Jaric (51) found that countermovement

jumps with zero load maximized mean power and jump height, yet the velocity (peak) of the center of mass at take-off increased by deloading bodyweight by 30% (51). Horizontally oriented exercises, including a novel exercise known as an assisted horizontal squat jump (Figure 11), have been shown to be beneficial to improving movement velocity due to the extremely high velocity reached by pushing against almost zero gravity (42,77). This exercise, part of a longitudinal training intervention aimed at improving F-v balance in individual profiles (44), was shown to produce extremely large changes in the velocity component of the F-v profile, as well as effecting increased jump heights.

Assisted sprinting may provide another unique approach for overloading the neuromuscular system at

Table 3 Exercises to improve maximal movement velocity	
Exercise	% 1RM/Load
Countermovement jump	BW
Assisted jumps	Assisting force to deload BW by 30%
Horizontal squat jump	<BW
Assisted horizontal squat jump	Assisting force 95–110 N
Squat jumps	BW—10% BW
Assisted sprinting	100–106% maximal velocity
Reactive jumps	BW
Box jump (bilateral and unilateral)	BW
Jump shrug	BW + 20–40 kg
Hang high pull	BW + 20–40 kg

BW = bodyweight; 1RM, 1 repetition maximum.

higher than the maximal voluntary velocities. Using a horizontal towing mechanism such as the DynaSpeed (MuscleLab) (Figure 12) or the 1080Sprint™, acute horizontal running velocities increased, along with lower limb electromyography activity, which suggested that higher neural activity took place with possible transfer to unassisted maximal sprinting

(54–57). However, given that maximal running velocity is by definition the far left side of the F-v spectrum, research should aim to verify whether training over an individual’s maximal voluntary running speed (i.e., overspeed training) benefits unassisted performances. Plyometric activities such as bounding, drop jumps, and reactive jumps are also recommended for athletes who want



Figure 10. Pre-load countermovement jump with band resistance, B: Toe-off position with band assistance (increased vertical velocity), C: Flight phase (increased jump height). Assisted vertical jumps to deload athlete bodyweight against gravity by 30% using an elastic band (51).

to improve force produced at high velocity due to the reliance on the stretch-shortening cycle (31). Table 4 identifies 2 resistance training sessions that could be performed across 1 week, which include both horizontally and vertically oriented exercises. These exercises and the associated sets, repetitions, and loads are programmed to improve the maximal movement velocity of the athlete.

IMPROVING MAXIMAL POWER (OPTIMAL LOADING CONDITIONS) FOR SPRINTING

Training at a load that is associated with movement velocity at which maximal mechanical power occurs has been shown to be the most effective method for increasing overall maximal power (36,46). Haff and Nimphius (36) define optimal load as that which maximizes mechanical power for a specific exercise. Within the strength and conditioning literature, the assessment of power is broad, with technology including force plates, linear position transducers (LPTs), and accelerometers, deriving power metrics that are used to determine optimal load. Therefore, the context of the variable must be understood and interpreted correctly when implementing into the training program (see the Limitations section). During resistance training, the optimal load for developing P_{MAX} in a jump squat has been shown to range from 0% of 1RM (18–20) to 30–45% of bench press 1RM in the bench press throw (68,82) and 70–80% of 1RM when performing weightlifting exercises such as the snatch and/or clean (18,22,46). This approach to training has also been used in cycling via torque-velocity tests to determine the optimal pedaling conditions (frequency) over a set distance (32). The discrepancy in power assessment and 1RM percentages across a range of exercises demonstrates a lack of clarity and inconsistency to determine the load to achieve P_{MAX}.

However, improving maximal horizontal power for sprinting requires focusing on the factors that contribute to this variable: horizontal force and horizontal



Figure 11. A: Loading against box in start position with band resistance, B: Toe-off position with band assistance (increased horizontal velocity). Assisted horizontal squat jump using a roller board and elastic band to push against reduced gravity (1,43,44).

velocity. Depending on the training phase and needs of the athlete, exercise prescription should be directed to all parts of the F-v spectrum to ensure that a strong level of mechanical effectiveness is maintained and to achieve the highest ratio of horizontal force during acceleration. In regard to sprinting, this may entail using specific exercises for producing maximum power by training at an optimal load (27). Optimal load training has previously been shown to be more effective for improving dynamic athletic performance compared with other loading conditions (68,92). However, combining different resistance training loads to improve power and ballistic

performance has also been shown to be effective in many studies (17,37,45,52,88,89). With this in mind, across the F-v spectrum, exercises to improve maximal power could include resisted sprinting, sprinting at maximum speed, jump squats (trap bar) (Figure 13), plyometrics (horizontal bounding), and assisted sprinting (Figure 12). Highlighting exercises from all aspects of the F-v spectrum should result in athletes maintaining or raising their mechanical effectiveness with limited decline in either contributing variable. Although all loads across the F-v spectrum contribute to P_{MAX} , Table 5 identifies 2 resistance training



Figure 12. Assisted sprinting using the DynaSpeed (MuscleLab) to allow athletes sprint at supramaximal speed (54,56,57).

sessions that could be performed across 1 week, which include both horizontally and vertically oriented exercises. These exercises and the associated sets, repetitions, and loads are programmed to focus on maximizing power.

LIMITATIONS TO SPRINT PROFILING

Coaches need to carefully consider the implications of the training interventions they select to address characteristics of the F-v profile. Potential weaknesses should be addressed but not at the expense of building on athlete strengths. It follows that strength and conditioning coaches need to keep sight of their primary training goals and use sprint profiling as a monitoring and diagnostic tool, similar to testing hamstring strength or force plate analysis of vertical jump actions to assess benchmarks. Detraining qualities is a risk if too much time is being expended on weaknesses. For example, an athlete who produces maximal power with lower force values will result in decreases in velocity values if training focuses on those particular movements for an extended period. This may impact on an athlete's ability to produce power in situations where force is required at different magnitudes and velocity (36).

Recent research suggests that the selection of exercises based solely on dynamic correspondence and the force-vector theory (34) contains certain limitations. A thorough understanding of dynamic correspondence often means that the selection of specific exercises is narrow and predetermined. Contreras et al. (14) and, more recently, Loturco et al. (49) suggest that the force-vector theory should be the primary focus when selecting exercises to improve sprint acceleration and maximal velocity. The theory states that the force-vector, which comes into play when sprinting, occurs in the anteroposterior direction relative to the body, and therefore, the exercises selected must focus on producing horizontal force and provide necessary time for skill acquisition, which in turn should improve transfer to the performance to a greater degree than vertically oriented exercises (14). Currently, most resistance

Table 4
Force production at high velocity

Exercise	Week 1		Week 2		Week 3		Week 4	
	Volume	Load	Volume	Load	Volume	Load	Volume	Load
Day 1—horizontal orientation								
Assisted sprinting (DynaSpeed)	1 × 20, 30, 40 m (upright posture)	101% of training maximal velocity (flying run)	1 × 30, 30, 40 m (upright posture)	102% of training maximal velocity (flying run)	1 × 30, 40, 50 m (upright posture)	103% of training maximal velocity (flying run)	1 × 20, 30, 40 m (upright posture)	101% of training maximal velocity (flying run)
Maximal sprinting	3 × 40 m (20 m accel + 20 m fly)	BW	3 × 50 m (20 m accel + 30 m fly)	BW	3 × 60 m (30 m accel + 30 m fly)	BW	2 × 40 m (20 m accel + 20 m fly)	BW
Horizontal bounding (8 contacts/set)	3 × 8 cts	BW	4 × 8 cts	BW	5 × 8 cts	BW	3 × 8 cts	BW
Day 2—vertical orientation								
Reactive hurdle hops (5 contacts/set)	3 × 5 cts	BW	4 × 5 cts	BW	5 × 5 cts	BW	3 × 5 cts	BW
Band-assisted vertical jumps	2 × 5	Deload BW 30%	3 × 5	Deload BW 30%	4 × 5	Deload BW 30%	2 × 5	Deload BW 30%
Double-leg depth jump to box	2 × 6	BW	3 × 5	BW	3 × 6	BW	2 × 6	BW

accel = acceleration, BW = bodyweight, cts = contacts, fly = flying run.

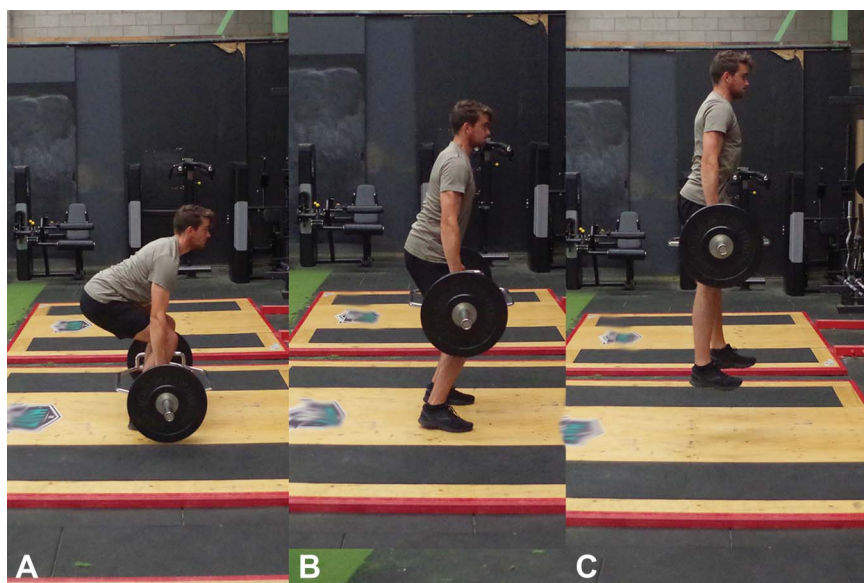


Figure 13. A: Start position of lift, B: Accelerating load vertically, C: Loaded flight phase. Jump squat with trap bar at optimal load (approximately 40% BW).

exercises are vertically oriented (72), for example, the back squat, deadlift, or weightlifting derivatives, therefore emphasizing vertical force production (72). However, this serves to negate horizontally oriented force and opposes dynamic correspondence and the force-vector theory. Notwithstanding, Loturco et al. (49) found strong correlations with sprint performance when performing hip extension-focused exercises, for example, hip thrust, with the initial phase of the sprint acceleration, whereas those exercises loaded vertically, for example, jump squat, showed greater transfer to the maximal velocity phase. Both of these findings, however, are in direct contrast to the findings of Seitz et al. (81) who found positive acceleration changes, 0–30 m, from studies primarily concerned with the back squat, and Jarvis et al. (41) who found no sprint performance transfer from an 8-week study using the hip thrust exercise. This suggests that conjecture remains in regard to the training axis, which should be used to enhance sprint acceleration performance.

Although the force-vector theory is intuitive in many respects, Fitzpatrick et al. (34) proposed that applying the force-vector theory to training was

a basic misunderstanding of simple mechanics. Primarily, the issue lies in understanding the difference between the direction of force relative to the global frame, as against the direction of force determined by the orientation of the athlete (35). This is evident in sprinting when the athlete adopts a triple flexion (front-side mechanics) position during acceleration and when reaching maximal velocity; the orientation of the body is at approximately 45° while accelerating and at approximately 90° when at maximal velocity. Kugler and Janshen (48) noted the strong relationship between body lean and direction of GRF, as leaning forward during acceleration places the athlete in an advantageous position for applying propulsive force. However, although orientation of force needs to be understood, from a practical standpoint, a combination of both vertically and horizontally loaded resistance training exercises seems to be the ideal approach when attempting to improve sprint performance (14,49,72,95).

Maximal power is a major performance indicator and thus frequently a priority when selecting exercises to improve dynamic, ballistic

performance (16,18,19,21,22,25,82,85,87), yet conjecture and difficulties exist when determining load for actions involving multiple joints. Maximal power and optimal load is influenced not only by the technology that derives the variable but also by whether it is specific to “system power” (external-whole body), joint power (internal—at a specific joint), or perhaps more applicable to the weight room, “bar power” (using an LPT). Previously, Cormie et al. (15) have recommended using a combination of a force plate and an LPT to best determine power in lower-body exercises. In a resistance training context, the discrepancies in how optimal load is reported therefore present a high level of ambiguity in reference to the load-power relationship and present issues with how practitioners can make sense of how to apply exercise prescription to not only improve maximal power but how optimal load is established (15). The use of a range of methodologies to determine power has led to a broad range of approximate 1RM percentages for which maximal power is developed in various exercises such as the power clean, squat, and jump squat. Considerations for effectively understanding and interpreting optimal load must also include the specific movement pattern(s) used, training history/status of the athlete, and whether the exercise uses single or multiple joints (22).

Cross et al. (29) noted that during unresisted sprint acceleration, maximal power was achieved within the first 2 seconds of the movement, and therefore, the remainder of the sprint occurred at a suboptimal load. To recreate and extend the conditions in which athletes move at an optimal load, resisted sprint training was introduced using a loaded sled that corresponded to approximately 96% of the athlete’s body mass or equivalent to a velocity decrement of ~50% of maximal velocity. This allowed athletes to sprint at optimal loads throughout the acceleration phase. In a later study, further changes to sprint acceleration

Table 5
Force production at maximal power (optimal load)

Exercise	Week 1		Week 2		Week 3		Week 4	
	Volume	Load	Volume	Load	Volume	Load	Volume	Load
Day 1—horizontal orientation								
Maximal sprinting	3 × 30 m (20 m accel + 10 m fly)	BW	4 × 30 m (20 m accel + 10 m fly)	BW	3 × 40 m (20 m accel + 20 m fly)	BW	2 × 40 m (20 m accel + 20 m fly)	BW
Resisted sprinting (DynaSpeed)	2 × 10, 20, 30 m	Individual P _{MAX}	3 × 20, 30 m	Individual P _{MAX}	2 × 3 × 30 m	Individual P _{MAX}	4 × 30 m	Individual P _{MAX}
Hip thrust	3 × 5	Individual P _{MAX}	3 × 5	Individual P _{MAX}	4 × 5	Individual P _{MAX}	3 × 5	Individual P _{MAX}
Day 2—vertical orientation								
Power snatch	4 × 3	30–50%	4 × 3	30–50%	5 × 2	30–50%	4 × 2	30–50%
Power clean	4 × 3	70–80%	4 × 4	70–80%	4 × 4	70–80%	5 × 2	70–80%
Jump squat (trap bar)	5 × 3	20–50%	5 × 3	20–50%	6 × 2	20–50%	5 × 2	20–50%
accel = acceleration, BW = bodyweight, fly = flying run, P _{MAX} = maximal power, % based off 1-repetition maximum in exercise.								

were not evident when athletes performed a resisted sprinting protocol at optimal load compared with groups that used lighter or heavier loads (29). However, the authors acknowledged that the current state of the individual F-v profile and the random group assignment may well have affected greater adaptations to sprint acceleration, and greater research is required.

One limitation recently noted on sprint profiling methodology using the “simple method” (63) is that the power variable only represents step-averaged external power produced in the horizontal direction to accelerate the body center of mass, neglecting the internal “joint” power (P_{int}) required to move the limbs around the center of mass (69). Pavei et al. (69) suggest that other mechanical components aside from horizontal power are needed to accelerate in a sprint such as the body center of mass and internal power (P_{int}). Although the simple method provides valuable insight into power in the horizontal direction across the sprint acceleration, there must be the understanding that no internal power variables are measured, and therefore, the overall power output computed via the simple method will be an underestimation of the total power developed by muscles but will rather characterize the power capabilities of overall sprint propulsion. Notwithstanding, the practical application to coaches using P_{int} is limited, considering the exhaustive technology necessary to obtain the data, and therefore, the simple method may be a more appropriate measure in the field. In addition, it is not known whether P_{int} is a performance indicator, thus a key variable of interest in training.

Therefore, when using optimal load as a training strategy to improve maximal power, it is prudent to understand the context of power being measured, along with incorporating a variety of loads across the F-v spectrum to ensure a balanced approach for force and velocity adaptations (25,88,89).

FURTHER CONSIDERATIONS

Most literature on sprint profiling discusses factors that contribute to the overall mechanical output across the performance; however, there has been a growing level of interest in understanding the application of sprint profiling in the rehabilitation field and return-to-play (RTP) protocols from hamstring injuries (53). Although not the primary focus of this review, the application of using mechanical variables of pre-injury performance and using these in the RTP protocols with sports medicine staff may provide further comparative data to ensure a safe return to performance. Mendiguchia et al. (53) identified that sprint profiling highlights the capability to produce horizontal force at low speed is a limiting factor to performance when returning from a hamstring injury; therefore, the application of sprint profiling as a monitoring tool to assess how force production changes across a competitive season or in response to an injury could be useful individual information to sports medicine staff.

SUMMARY

Sprint profiling using the field methods briefly outlined in this review offers an innovative and alternative approach to understand the mechanical determinants of sprint acceleration. Although further research and experimental evidence is needed, together with applied longitudinal exercise interventions, the field method is a practical and valid approach that allows strength and conditioning coaches to access kinetic data on sprint acceleration, which previously was only attainable in a laboratory. These data allow coaches to design individualized training programs.

The resistance training program used to address mechanical effectiveness should consist of exercises that focus on both horizontal and vertical force production, acknowledging the limitations to the force-vector theory; however, a priority could be placed on one orientation over the other depending on the phase of the training cycle or the needs of the athlete. Sprint profiling can be used for athletes involved in

sports where sprint acceleration is crucial and for identifying and changing the variables contributing to performance. It may allow coaches to devise individualized training programs to a greater degree compared with traditional methods as a means of enhancing sprint acceleration and improving the effectiveness of force application.

Guidelines for implementing a training and/or rehabilitation program that addresses the mechanical variables of horizontal force-velocity and power-velocity include the following:

- Assess the capabilities of producing horizontal force and the mechanical effectiveness of force application during sprint acceleration (sprint profile)
- Identify any existing F-v imbalance across sprint acceleration.
- Prescribe appropriate training programs to address the needs of the athlete and the slope of the profile.
- Reassess the athlete after an appropriate period to determine adaptations to mechanical effectiveness and changes to sprint acceleration.

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Dylan Shaun Hicks is a PhD candidate and teaching faculty member in Exercise Science at Flinders University, South Australia.



Jake George Schuster is Senior Sports Scientist at Vald Performance and Affiliate Researcher at Florida State University's Institute of Sports

Science and Sports Medicine.



Pierre Samozino is Associate Professor in the Sport Science department and the Laboratoire Interuniversitaire de Biologie de la Motricité at the University Savoie Mont Blanc.



Jean-Benoit Morin is a Full Professor at the Université Côte d'Azur, Director of the Masters in Sports Performance & Training Science and the Associate Dean for Research, Faculty of Sport Science.

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